Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2025
Team ID	GACSE-114
Project Name	FitBuddy – AI-Powered Fitness Planner
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Fitness Profile Input	User enters name, age, weight, fitness goal, and workout intensity Input form collects user fitness data Data validation for required fields
FR-2	AI Workout Plan Generation	Generate personalized 7-day workout plan Use AI model to create structured exercise routines Display generated plan to the user
FR-3	Workout Plan Feedback Update	User submits feedback about generated plan System processes feedback using AI Updated workout plan is generated
FR-4	Workout Plan Storage	Store user fitness details in database Store generated workout plans Retrieve stored plans for viewing
FR-5	Nutrition & Recovery Guidance	Provide goal-specific nutrition tips Provide recovery recommendations Display health guidance with workout plan
FR-6	Admin View / Plan Management	View all users and generated plans Display stored workout records Manage stored user fitness data

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The application must provide a simple and intuitive web interface that allows users to easily input fitness information and view generated workout plans.
NFR-2	**Security**	Sensitive configuration data such as API keys must be securely stored using environment variables to prevent unauthorized access.
NFR-3	**Reliability**	The system must reliably generate workout plans and handle user requests without errors or system failures.

NFR-4	**Performance**	The application should generate workout plans quickly with minimal response time while efficiently handling API calls and database operations.
NFR-5	**Availability**	The application should remain accessible through a web browser whenever the server is running.
NFR-6	**Scalability**	The system architecture should support future enhancements such as mobile applications, wearable integration, and additional AI features.